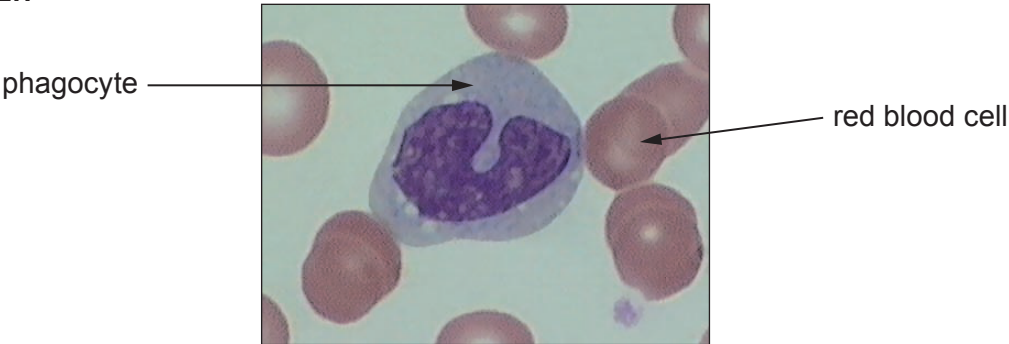


2. **Image 2.1** shows a sample of human blood as seen through a light microscope.

Image 2.1

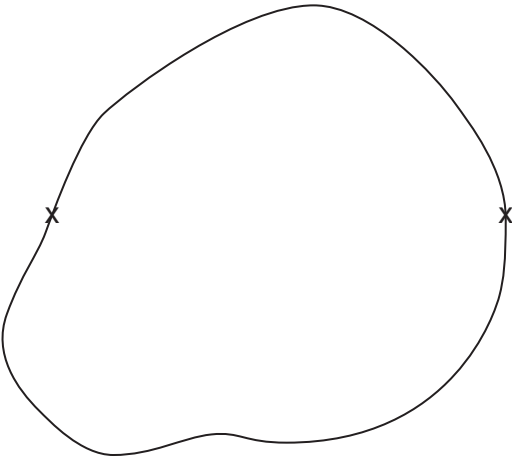


(a) (i) The drawing in **Image 2.2** shows the cell membrane of the phagocyte.

I. **Draw** the **nucleus** in the outline of the phagocyte and **label** it. [1]

II. **Label** the **cytoplasm** in the outline of the phagocyte. [1]

Image 2.2



(ii) I. Measure the width x-x **in mm** on the drawing. [1]

Width x-x = mm

II. The width of the actual cell is 0.012 mm.
Calculate the magnification of the drawing using the equation below. [1]

magnification = $\frac{\text{width x-x on drawing}}{\text{width of the actual cell}}$

magnification = ×



(b) **Complete the table** by selecting functions from those below.

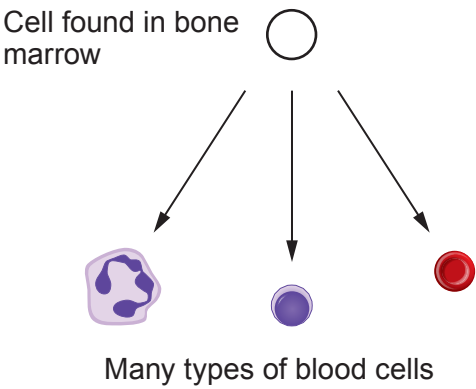
[3]

- transports hormones
- carries oxygen
- defends against disease
- clots the blood
- controls temperature

Part of blood	Function
red blood cell	
white blood cell	
platelets	

(c) There are many types of blood cell, each with a particular function. They all come from one type of cell found in bone marrow through the process of differentiation. This is shown in **Image 2.3**.

Image 2.3



Choose the letter **A-D** to give the term used for the various types of blood cells.

[1]

- A selected
- B specialised
- C secondary
- D sensitive

Answer letter =

